GAYATHRI P

Associate Data Engineer

Cognizant Technology solutions

gayathri4198@gmail.com

Bengaluru, India 9916493252

linkedin.com/in/gayathri-p-78283817a

Data Engineer with 4+ years of experience in building and optimizing large-scale data pipelines, distributed processing frameworks, and cloud migration solutions. Proficient in Python, PySpark, Scala, and Big Data platforms (Hadoop, Spark, Hive) with proven expertise in AWS-based data engineering (S3, EMR, Glue, Athena, Redshift). Adept at coding, performance tuning, data transformation, orchestration, and production support across batch and streaming workloads. Focused on scalable, maintainable, and high-performing data solutions.

Skills

- Programming & Big Data:Python, PySpark, Scala, Apache Spark, Spark SQL, Hadoop (Cloudera 3.5 yrs, Hortonworks 1 yr), Databricks (basic exposure), HDFS, Hive, HBase, Sqoop, ORC, JSON, CSV
- Cloud & Data Platforms:AWS (S3, EMR, Glue, Athena, Redshift, DynamoDB, Lake Formation), Apache Airflow
- ETL & Orchestration:Tidal Automation, Workflow Scheduling, Job Dependencies, Alert Monitoring
- Databases & Querying:Oracle, MySQL, Hive, SQL
- Data Modeling & Optimization: Hive Table Design, Dynamic Partitioning, Query Tuning, Spark Job Optimization, Data Validation
- Streaming & Production Support: Kafka, Spark Streaming, Pega SWC, VOC, SLA Monitoring, Incident Resolution
- Version Control & Scripting: Git / Bitbucket, Shell Scripting, UNIX/Linux

Professional Experience

Bengaluru| Dec 2020 - Jan 2025

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Client: Charter Communications(Spectrum Data Lake)

- Contributed to end-to-end migration of on-prem Hadoop workflows to AWS using S3, Glue, EMR, Athena, and Redshift, ensuring scalable, cost-effective architecture.
- Developed and optimized PySpark and Scala-based ETL pipelines for structured and unstructured VOD data ingestion from legacy systems (TWC, BHN, Charter).
- Implemented Spark job tuning techniques (caching, partitioning, broadcast joins) to improve performance by 25% and reduce job runtimes.
- Integrated and orchestrated workflows using Tidal Automation, managing job dependencies, error recovery, and alert-based monitoring.
- Supported real-time streaming pipelines using Kafka, Spark Streaming, Pega SWC, and VOC, improving event processing efficiency and reducing reporting delays by 15%.
- Designed data validation and transformation frameworks, ensuring consistency across CX and AM applications; delivered ORC-based outputs for downstream analytics.
- Participated in AWS cloud enablement — configuring Glue jobs, EMR clusters, and Athena queries for analytics.
- Collaborated with cross-functional teams to debug, document, and enhance existing data pipelines and mentored junior engineers to strengthen coding best practices.

Roles and Responsibilities | Charter Communications (Spectrum Data Lake)

- Designed and optimized ETL pipelines in PySpark, Spark SQL, and Hadoop for ingesting VOD and CX datasets, enhancing data accessibility by 20%.
- Implemented API-based data integrations and automated transformations for multi-source ingestion workflows.
- Deployed alert-driven monitoring for real-time data processing using Kafka and Pega SWC, ensuring zero SLA breaches.

- Engaged in data modeling, schema management, and query optimization to improve reliability of analytics dashboards.
- Ensured robust version control, unit testing, and CI/CD compliance with Git and Tidal.
- Delivered production-ready code aligned with enterprise-grade quality and coding standards.

Project Highlights

- AWS Migration Enablement: Modernized on-prem Hadoop pipelines into AWS-native solutions using EMR, Glue, and Athena.
- Spark Performance Optimization: Improved Spark job efficiency by optimizing shuffle operations, join strategies, and file formats.
- Streaming Workflow Enhancement: Enhanced Kafka-based message pipelines for faster VOC insights and reduced data latency.

Certifications/Training

- Apache Hadoop & Spark (Udemy)
- PySpark Development (Udemy)
- SQL Developer Certification (Udemy)
- Unix Fundamentals (Udemy)
- Leadership Development Program – Cognizant
- Professional Training: Completed modules in PySpark & AWS Data Engineering via Scaler online training program
- AWS Practitioner certificate - AWS
- AWS Certified Data Engineer – Associate (In Progress)

Education

B.Tech in Computer Science and Engineering
GITAM Deemed to be University | 2016 – 2020
CGPA: 7.85/10

Additional Information

- Languages: English, Telugu, Kannada
- Bluebolt Ideas: Submitted 3+ ideas, improving automation efficiency by 10%
- Interviewing & Mentorship: Conducted 5+ interviews and mentored 5+ juniors with 100% onboarding success
- Enthusiastic about continuous learning, code optimization, and cloud-native architecture.
- Open to roles involving big data development, pipeline engineering, and AWS data platform modernization.